Total No. of Questions : 8]

SEAT No. :

PB2539

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B.E. (Electronics & Telecommunication)/(Honors in Robotics)

ARTIFICIAL INTELLIGENCE IN ROBOTICS

(2019 Pattern) (Semester - VIII) (404183 HR)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume Suitable data if necessary.*

Q1) a) Explain Hill climbing algorithm. Explain Local maxima, Global Maxima and Plateau. **[8]**

b) Differentiate between uninformed and informed search methods. **[5]**

c) One aspect of a simulated annealing cooling schedule is the temperature. How do you decide on a suitable starting temperature? **[5]**

OR

Q2) a) Describe the Tabu search algorithm. Justify, Tabu search cannot guarantee the optimal result. **[8]**

b) What is the difference between heuristic and metaheuristic? **[5]**

c) Why are real-coded genetic algorithms necessary? **[5]**

Q3) a) Describe the Canny edge detection technique with the application. **[7]**

b) What is imaging-based automatic sorting? **[5]**

c) Explain two object recognition techniques used in industries. **[5]**

OR

P.T.O.

Q4) a) Explain the segmentation methods used in the vision system with a suitable example. **[7]**

b) Enlist the applications of a machine vision system and explain any one application of machine vision system in manufacturing. **[5]**

c) What is imaging based automatic inspection? **[5]**

Q5) a) What are the applications of intelligent systems for mobile Robot Motion Planning? **[8]**

b) Explain path planning robot control in dynamic environments. **[5]**

c) Explain any one application of intelligent robot. **[5]**

OR

Q6) a) What is Task based Hybrid Closure Grasping Optimization for Autonomous Robot Arm, explain with example. **[8]**

b) How intelligent robot maintain accurate motion. **[5]**

c) Enlist advantages and limitations of intelligent robot. **[5]**

Q7) a) Explain route optimization for AS/RS systems. **[7]**

b) Comment on how robotics can be used to design intelligent vehicles. **[5]**

c) How real time scheduling is done using AI. **[5]**

OR

Q8) a) Explain the role of artificial intelligence in flexible automation. **[7]**

b) Comment on the fundamental problem in robotics. **[5]**

c) Explain the process of resolution with a proper example. **[5]**

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